

MOHSINAT AHMED LABONI

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RESEARCHGATE	researchgate.net/profile/Mohsinat-Laboni	WEBSITE	mohsinatahmedlaboni.github.io

EDUCATIONAL BACKGROUND

B.Sc. in Civil Engineering

Jun 2021 – Oct 2025

(Specialization: Environmental Engineering)

Islamic University of Technology (IUT), Bangladesh

CGPA: 3.65/4.00 (Credits: 181)

Thesis: Evaluation of Chemical and Natural Filtration Media for Microplastic Removal from Surface Water Following Coagulation–Flocculation Treatment

Final Year Design Project: Integrated Structural, Water, and Urban Infrastructure Design for a Multi-story Residential Building in Jamalpur Sadar

Additional Research Support:

- Provided research support for an M.Sc. thesis project
- Assisted in developing the experimental design framework and conducted the full laboratory workflow, from field sampling to microplastic quantification following NOAA protocols, and characterization through optical microscopy, SEM, and FT-IR

WORK EXPERIENCE

Research Assistant – University of Washington (UW), Bangladesh University of Engineering & Technology (BUET) June 2026 – Present

| Dhaka, Bangladesh

- Contributing to the development of a high-resolution national emission inventory for Bangladesh.
- Supporting reduced-complexity and empirical air pollution modeling for Bangladesh.
- Assisting with PM_{2.5} monitoring, chemical speciation, and source apportionment studies.
- Processing and analyzing air quality, meteorological, satellite, and GIS datasets.
- Contributing to research outputs supporting evidence-based air quality management and policy development.

Research Officer – CarboBon (carbobon.com)

Nov 2025 – May 2026

| Dhaka, Bangladesh

- Conducting carbon emissions analysis in line with GHG Protocol, IPCC guidelines, and ISO 14064 pre- and post- interventions based on ESG and capacity building initiatives.
- Participation in KII and FGDs and factory visits for on-site data verification
- Support carbon projects in biochar, sustainable agriculture, and biogas
- Contribute to field research and digital MRV tool development

Outside Plant (OSP) Design Engineer – Skarion (skarion.com), Inuberry Global (inuberry.com)

Feb 26 – Present

| Fairfax, Virginia, USA

- Worked in High-Level Design (HLD) using AutoCAD and GIS tools
- Contributed in route planning, network layout design, and documentation for telecommunication infrastructure

Research and Feature Journalist – The Financial Express Bangladesh Aug 2022 – Dec 2023
(thefinancialexpress.com.bd), Industry Insider (industryinsiderbd.com)

| Dhaka, Bangladesh

- Researched, conducted in-depth interviews with professionals and experts, and authored feature articles on education, engineering, and cultural affairs.

PROJECTS

Promoting Green Growth in the Ready-Made Garments Sector through Skills (PROGRESS)

(Details of the project can be found [here](#))

- Managed decarbonization data for 15+ RMG factories under the PROGRESS project, validating 8.46M kgCO₂e in emission reductions, with two factories achieving 19–43.5% absolute emission reductions.
- Developed technical roadmaps for GHG Protocol and ISO-compliant reporting, mapping 37 unique interventions and overseeing data integrity for a portfolio with 7.32M kgCO₂e in attributed savings.

Holly Springs Fiber Optics Internet Project

- Designed 400,000+ ft of XGS-PON networks, creating 100+ permit packages and BOMs,
- Supported \$2.8M+ in procurement. Designed underground and aerial infrastructure using ArcGIS Pro and AutoCAD in accordance with NESC/NEC standards. Led 500+ residential fiber deployments with HLD/LLD.

Greensboro Fiber Optics Internet Project

- Developed BOQs and AutoCAD construction drawings for the Greensboro (GB) fiber expansion, covering 170,000+ ft of underground conduit and 2,000+ residential deployments, totaling ~\$1.7M in OSP procurement.

INDUSTRIAL TRAINING

Academic Trinee, Public Works Department (PWD), Bangladesh

2 Jun 2022

– 27 Jun 2022

- Gained hands-on experience in civil engineering practices, including site inspection, project planning, and construction management.
- Learned about structural designs and analysis and understanding material selection all in compliance with BNBC 2020, ACI guidelines, including quality control procedures.

PUBLICATIONS

1. **Laboni, M. A.**, Tashfiah, R., & Ahsan, A. (2024). Study of yearly variation of water quality parameters and indices of the Turag River of Bangladesh (Paper ID 400). In Proceedings of the 7th International Conference on Advances in Civil Engineering. Volume, Issue and Pages: 3316-3323;
researchgate.net/publication/397638564_Study_of_yearly_variation_of_water_quality_parameters_and_indices_of_the_Turag_river_of_Bangladesh

2. **Laboni, M. A.**, Tashfiah, R., & Ahsan, A. (2025). A data-driven approach to PM_{2.5} prediction in Dhaka using machine learning techniques (Paper ID 634). In Proceedings of the International Conference on Civil Engineering Research & Innovations. Volume, Issue and Pages: 2891-2896; researchgate.net/publication/399962032_A_Data-Driven_Approach_to_PM25_Prediction_in_Dhaka_using_Machine_Learning_Techniques

WORK IN PROGRESS

1. **Laboni, M. A.**, Prantor, R. M., Karim, M. R., Islam, M. S., Baki, A. B. M., Sadiqa, W., & Nabila, A. T., Andaluri G. (2025). Predicting and Enhancing Microplastics Removal: A Data-Driven Mixture-of-Experts Approach for Coagulant-Specific Performance (*submitted for journal publication in 2025*)

2. Rimi, N.A., Prantor, R.M., Absar N.M., **Laboni, M.A.**, Fariha, Z., Sharaf, K. (2026). Rights and realities: Evaluating WASH inequalities in slums across Dhaka's two city corporations. International Conference on Climate Extremes, Disasters, Anticipatory Actions and Human Health (CEDAAH-2026)

3. Microplastic abundance in urban road dust across Dhaka City, Bangladesh: A probabilistic modelling approach (*expected submission - 2026*)

LANGUAGE PROFICIENCY

English: C1 (IELTS: 8.0 – Listening: 8.0, Reading: 8.0, Writing: 7.5, Speaking: 8.0; TRF: 25BD502896LABM001A)

EXTRA-CURRICULAR ACTIVITIES

- **Team Leader (Marketing) at AIESEC in Bangladesh**
 - Led marketing campaigns and team operations at university level, creating digital content and organizing outreach events to increase engagement and recruitment.
- **Assistant Head (PR & Communications) at ACI IUT Student Chapter**
 - Managed social media accounts and created contents to promote the chapter events in order to increase visibility and engagement
 - Coordinated publicity for workshops, seminars, and student activities.
- **Head of Library & Resources at IUT Al Biruni Society of Scientific Studies**
 - Facilitated workshops and study sessions to support members' academic and research activities
 - Mentored junior members in resource management and research support processes.

MEMBERSHIP OF PROFESSIONAL BODIES

- American Concrete Institute (ACI)
- Institute of Transportation Engineers (ITE)

CERTIFICATIONS

- Introduction to Image Processing, MathWorks

- Climate Change: Mitigation and Adaptation, Council on Foreign Relations
- ESG and Climate Change, University of Pennsylvania
- Climate Change: Carbon Capture and Storage, The University of Edinburgh
- GIS, Mapping, and Spatial Analysis (Specialization), University of Toronto
- Python for Data Science, AI & Development, IBM

TECHNICAL SKILLS

Environmental modelling and analysis	EPA SWMM, EPANET, Etank, Bioscreen
GIS software	ArcGIS Pro
Structural analysis and design	AutoCAD, ETABS, SAP2000
Programming and Data Analysis	Python, MATLAB
Statistical modelling	Design-Expert, MS Excel
Documentation	MS Word, Latex, Google Docs
Spreadsheets	MS Excel, Google Sheets
Presentation	MS PowerPoint, Google Slides, Canva
Graphic design	Adobe Photoshop, Canva
Video-editing	Sony Vegas Pro, Adobe After Effects

RECOMMENDATIONS

Sheikh Mokhlesur Rahman, Ph.D.
[Google Scholar Profile](#)
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Provat Kumar Saha
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